



Service Tool Instruction

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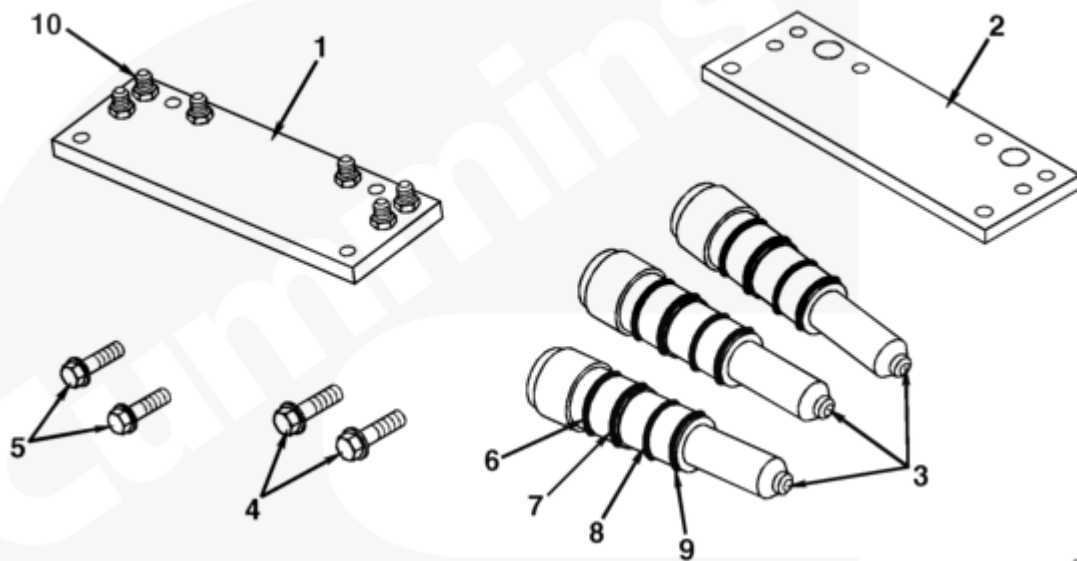
Description

Cylinder Head Fuel Passage Leak Test Kit

Purpose

This document provides information for using cylinder head fuel passage leak test kit, Part Number 3164943. The kit is used to leak test the fuel passages in the cylinder head on the Signature, ISX, ISX with CM870, and QSX15 engines.

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Table 1, Cylinder Head Fuel Passage Leak Test Kit, Part Number 3164943

Item Number	Part Number	Description	Quantity
1	3164945	Plate	1
2	3164946	Gasket	1
3	3164923	Injector cavity plug	3
4	3089232	Hexagon flange head capscrew (long)	2
5	3089016	Hexagon flange head capscrew (short)	2
6	3330537	O-ring seal	3

Table 1, Cylinder Head Fuel Passage Leak Test Kit, Part Number 3164943

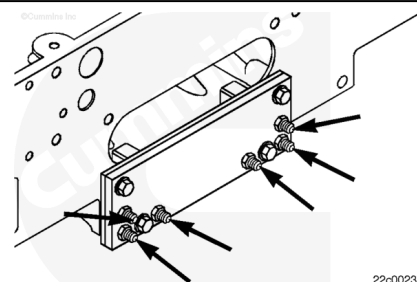
Item Number	Part Number	Description	Quantity
7	3330538	O-ring seal	3
8	3347937	O-ring seal	3
9	3347939	O-ring seal	3
10	68138	Male connector	6

Table 2, Item Used with the Cylinder Head Fuel Passage Leak Test Kit, Purchased Separately

Item Number	Part Number	Description	Quantity
1	3164231	Leak test kit	1

The fuel system is split into two banks. The front bank is for cylinders one through three, and the rear bank is for cylinders four through six. For this configuration, the banks are tested independently of each other.

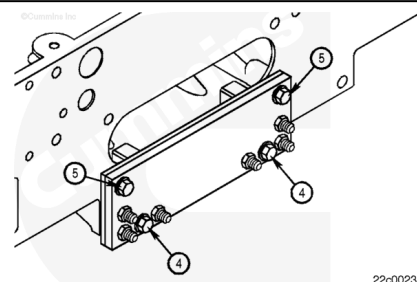
The left three male connectors on the plate are for the front bank and the right three male connectors are for the rear bank.



Note : If the male connectors are not installed in the plate, install them now using thread sealant.

Install the gasket, plate, and the two short capscrews (5) and two long capscrews (4). Tighten the capscrews.

Torque Value: 27 n·m [20 ft-lb]



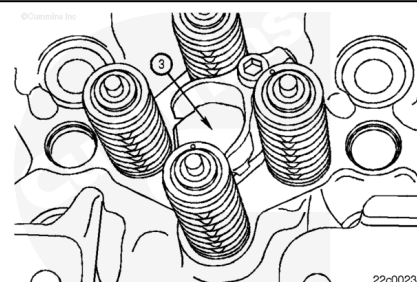
Inspect and lubricate the o-rings on the injector cavity plugs (3). Replace any worn or torn o-rings.

Install the three Injector cavity plugs (3) into either the front bank or the rear bank injector bores as required for testing.

Install the injector hold down clamps and capscrews.

Tighten the capscrews.

Torque Value: 41 n·m [30 ft-lb]



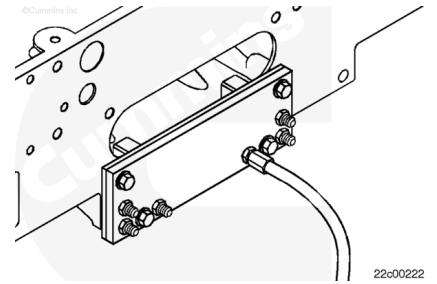
Attach the leak test kit, Part Number 3164231, or equivalent, to one of the three male connectors (metering, timing, or return) on the plate.

Apply 207 kPa [30 psi] of air pressure and shut off the air supply with the ball valve.

The fuel passage **must** maintain a minimum of 193 kPa [28 psi] for 30 seconds.

If 193 kPa [28 psi] is **not** maintained for 30 seconds, verify that none of the test hardware is leaking.

The **only** repair available is the replacement of the o-ring sealed plugs located at the ends of the cylinder head that terminate the fuel rail drilling. If the leaks are from any location other than these plugs, the cylinder head **must** be replaced.



Perform the same procedure for each of the other two passages on the bank of cylinders.

Remove the three injector cavity plugs (3) and move them to the other bank of cylinders as required for testing and repeat the above procedures.

